US012387437B2

(12) **United States Patent**
Guay et al.

(10) **Patent No.: US 12,387,437 B2**
(45) **Date of Patent: *Aug. 12, 2025**

(54) **AUTOMATED IMAGE AUGMENTATION USING A VIRTUAL CHARACTER**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/102,611**

(22) Filed: **Jan. 27, 2023**

(65) **Prior Publication Data**

US 2023/0177784 A1 Jun. 8, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/037,745, filed on Jul. 17, 2018, now Pat. No. 11,600,047.

(51) **Int. Cl.**
G06T 7/73 (2017.01)
G06T 19/00 (2011.01)

(52) **U.S. Cl.**
CPC ***G06T 19/006*** (2013.01); ***G06T 7/73*** (2017.01); *G06T 2207/30196* (2013.01)

(58) **Field of Classification Search**
CPC G06T 19/006; G06T 7/73; G06T 2207/30196

(Continued)

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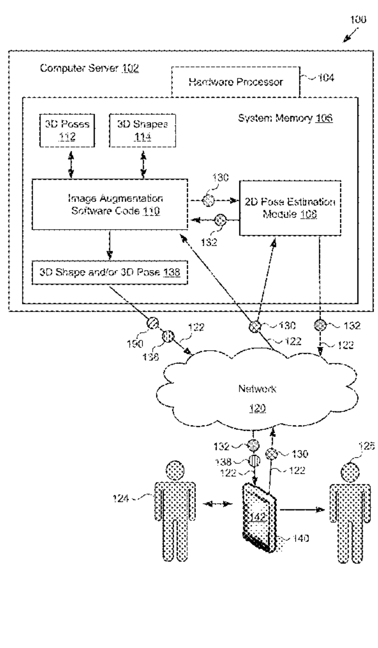
(74) *Attorney, Agent, or Firm* — Farjami & Farjami LLP

(57) **ABSTRACT**

An image processing system includes a computing platform having a hardware processor and a system memory storing an image augmentation software code, a three-dimensional (3D) shapes library, and/or a 3D poses library. The image processing system also includes a two-dimensional (2D) pose estimation module communicatively coupled to the image augmentation software code. The hardware processor executes the image augmentation software code to provide an image to the 2D pose estimation module and to receive a 2D pose data generated by the 2D pose estimation module based on the image. The image augmentation software code identifies a 3D shape and/or a 3D pose corresponding to the image using an optimization algorithm applied to the 2D pose data and one or both of the 3D poses library and the 3D shapes library, and may output the 3D shape and/or 3D pose to render an augmented image on a display.

20 Claims, 7 Drawing Sheets



(58) **Field of Classification Search**
USPC 345/419
See application file for complete search history.

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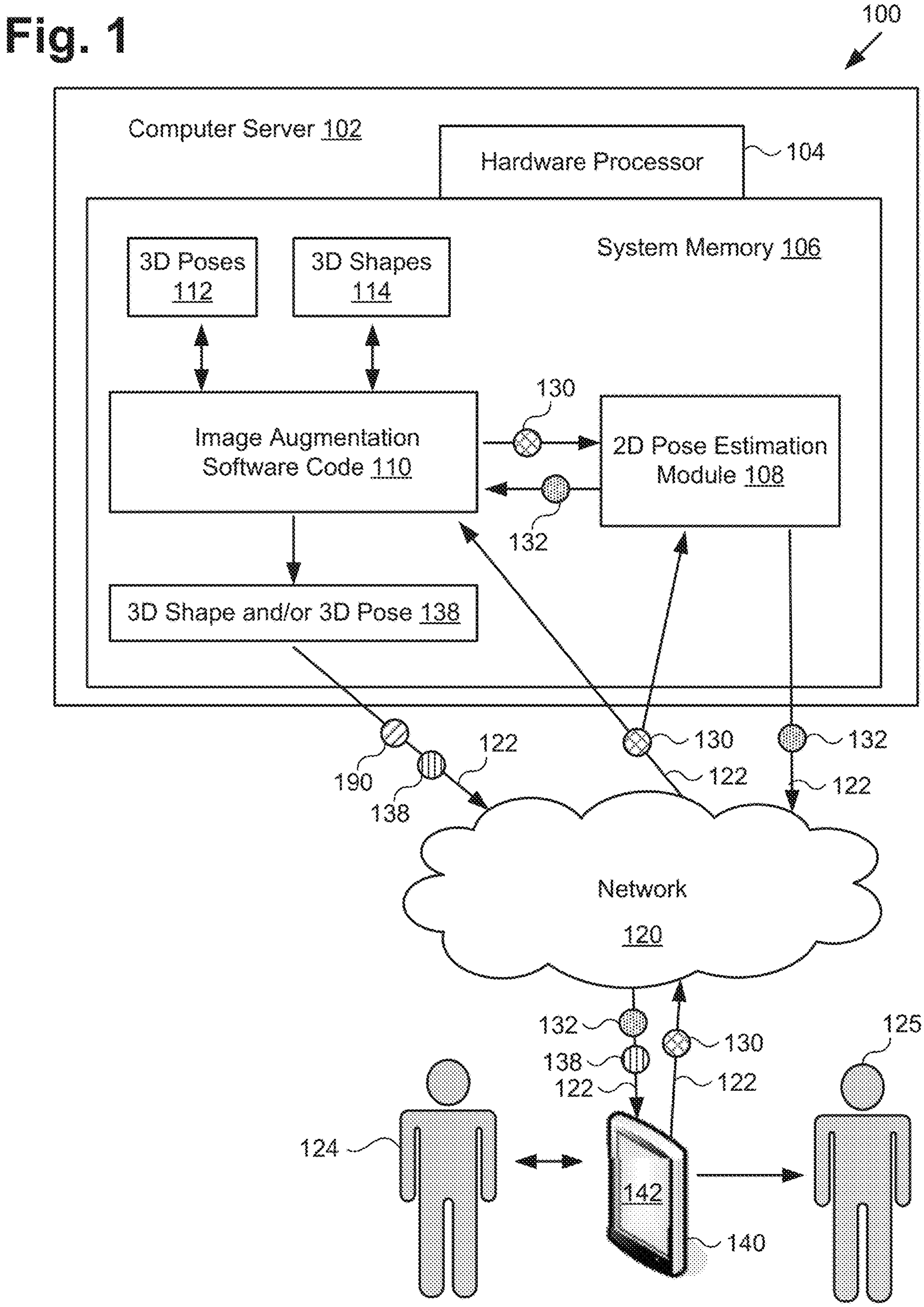


Fig. 1
100
Computer Server 102
Hardware Processor
104
System Memory 106
3D Poses
112
3D Shapes
114
Image Augmentation
Software Code 110
130
132
2D Pose Estimation
Module 108
3D Shape and/or 3D Pose 138
190
138
122
130
122
132
122
Network
120
132
138
122
130
122
125
124
142
140

Fig. 2

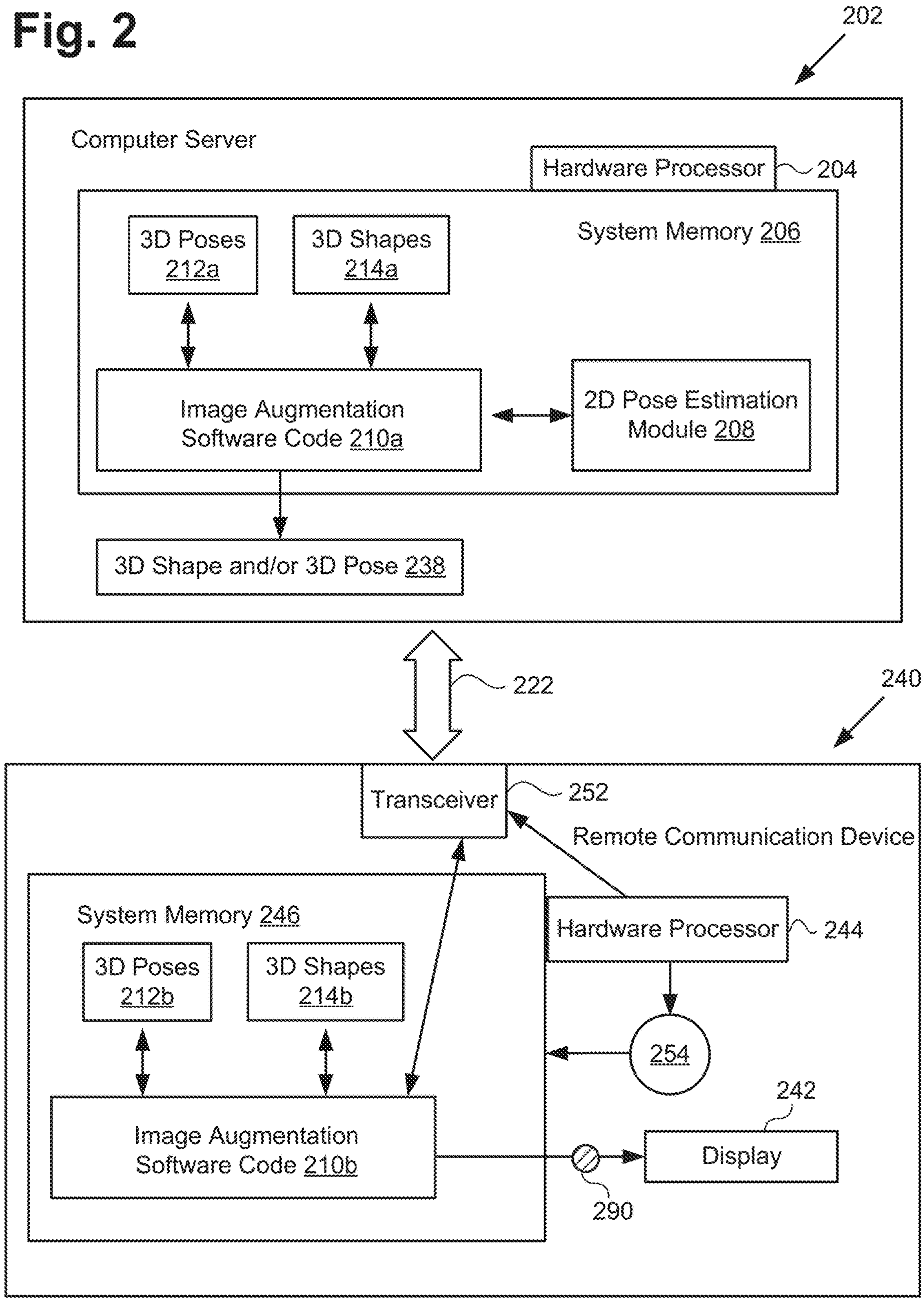

202
Computer Server
Hardware Processor
204
3D Poses 212a
3D Shapes 214a
System Memory 206
Image Augmentation Software Code 210a
2D Pose Estimation Module 208
3D Shape and/or 3D Pose 238
222
240
Transceiver
252
Remote Communication Device
System Memory 246
Hardware Processor
244
3D Poses 212b
3D Shapes 214b
254
242
Image Augmentation Software Code 210b
Display
290

Fig. 3

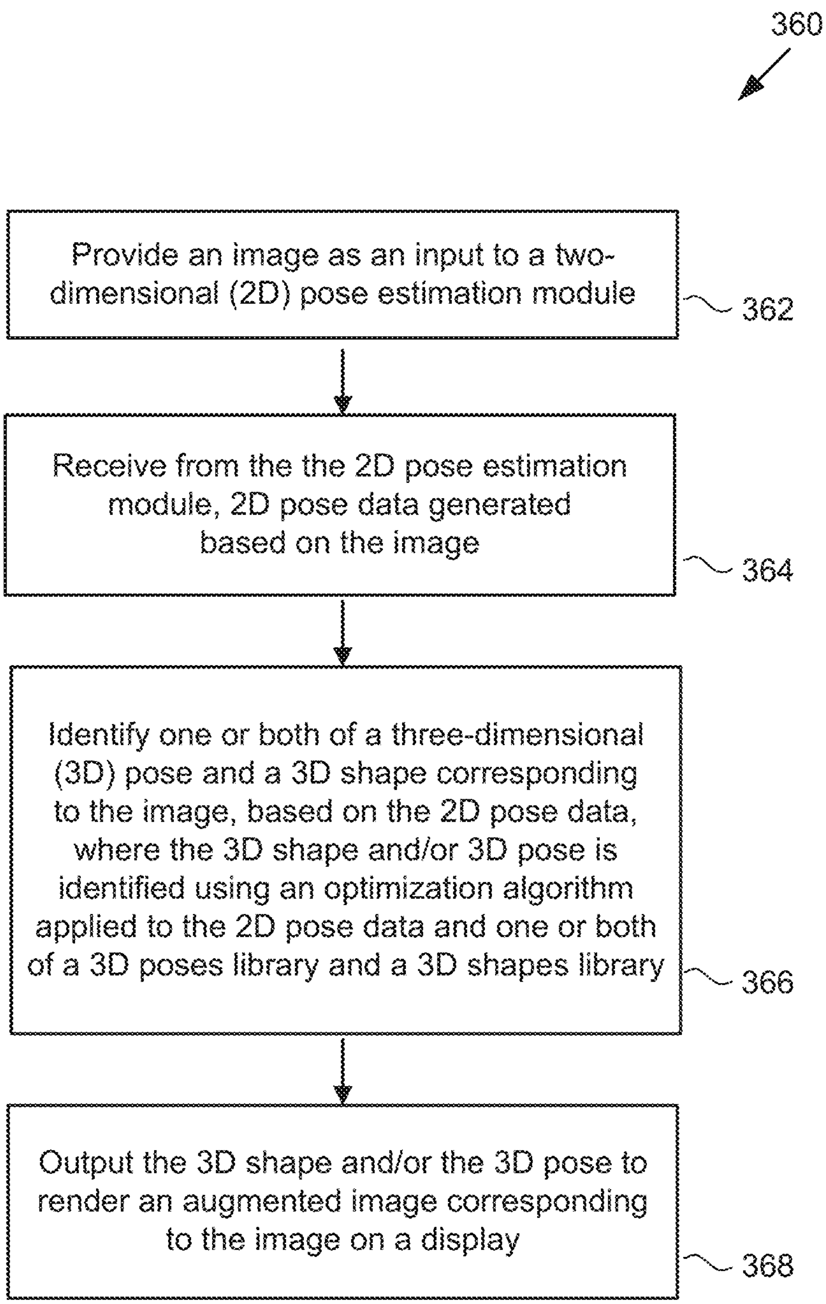
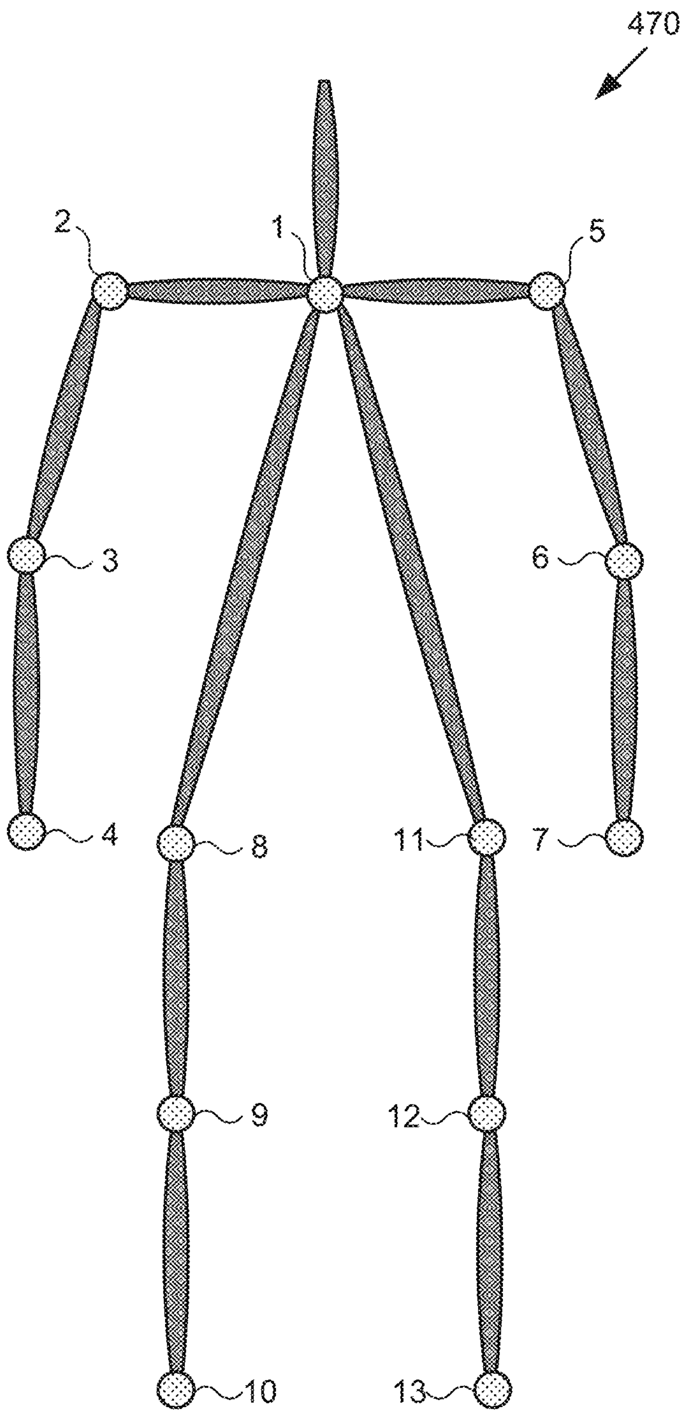

360
Provide an image as an input to a two-dimensional (2D) pose estimation module
362
Receive from the the 2D pose estimation module, 2D pose data generated based on the image
364
Identify one or both of a three-dimensional (3D) pose and a 3D shape corresponding to the image, based on the 2D pose data, where the 3D shape and/or 3D pose is identified using an optimization algorithm applied to the 2D pose data and one or both of a 3D poses library and a 3D shapes library
366
Output the 3D shape and/or the 3D pose to render an augmented image corresponding to the image on a display
368

Fig. 4



1. Neck
2. Right Shoulder
3. Right Elbow
4. Right Wrist
5. Left Shoulder
6. Left Elbow
7. Left Wrist
8. Right Hip
9. Right Knee
10. Right Ankle
11. Left Hip
12. Left Knee
13. Left Ankle

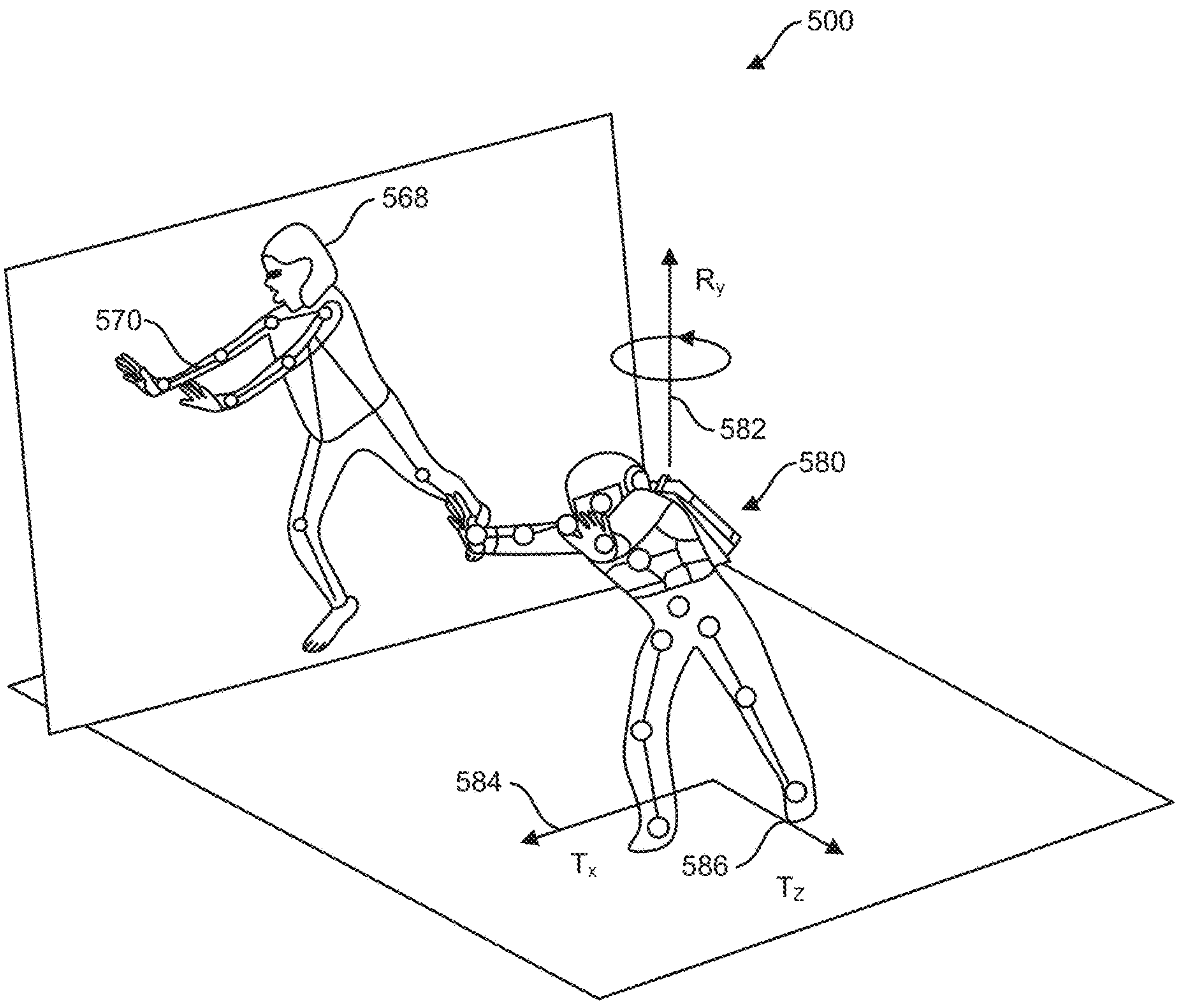


FIG. 5

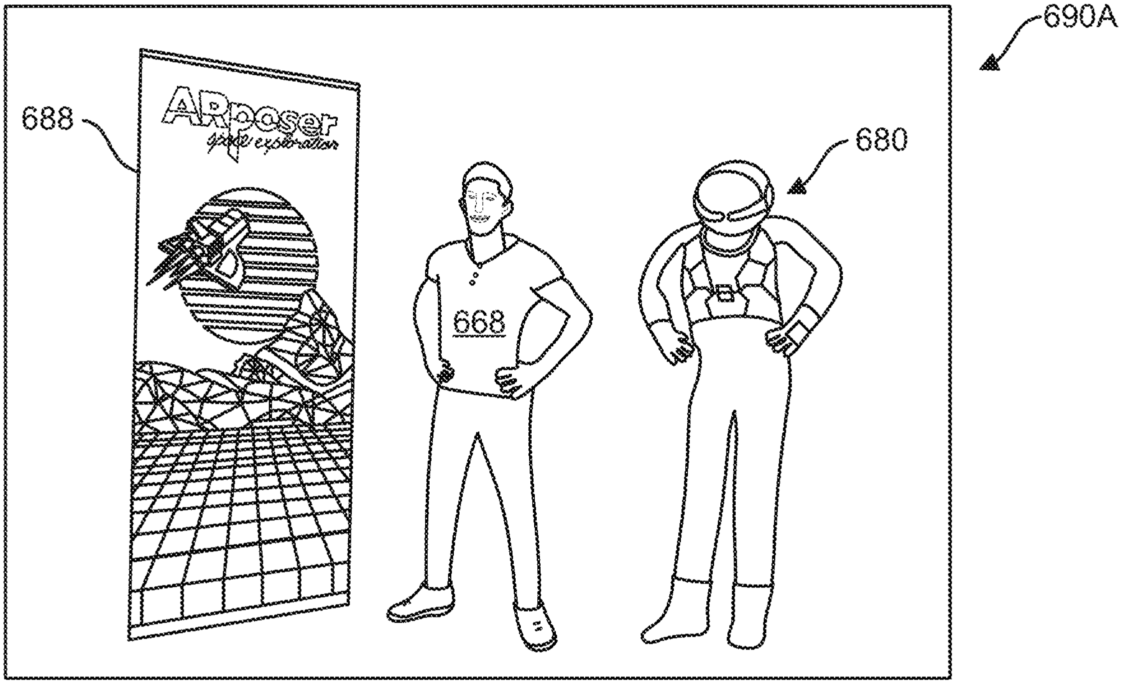

690A
688
ARposer
space exploration
680
668

FIG. 6A

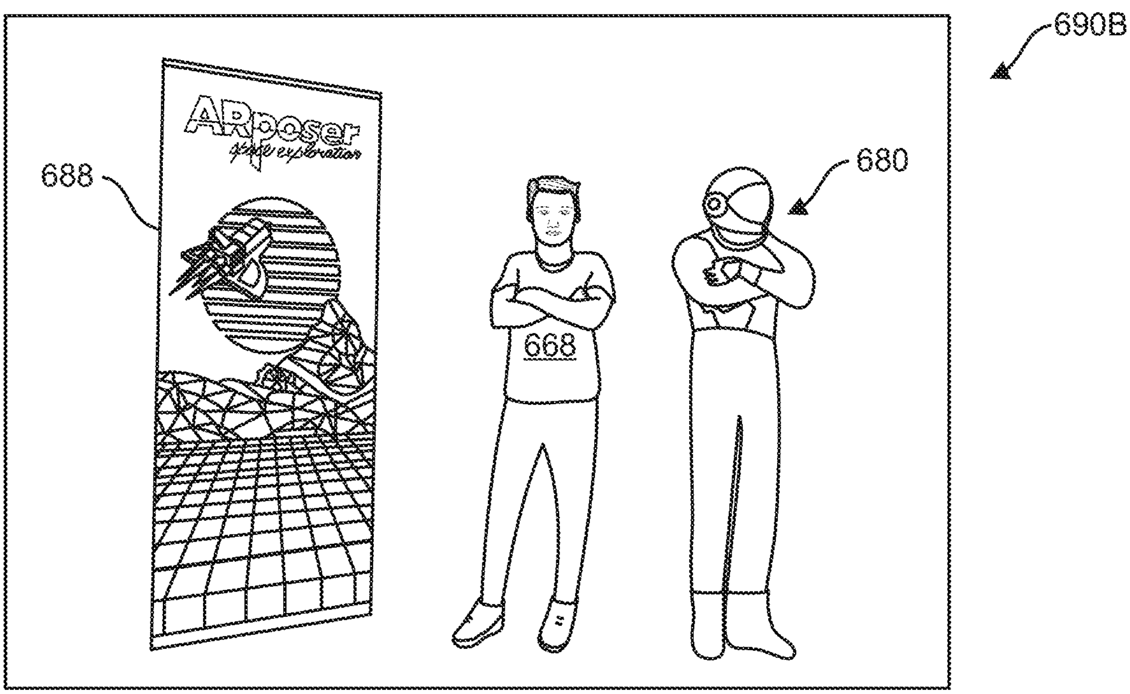

690B
688
ARposer
space exploration
680
668

FIG. 6B

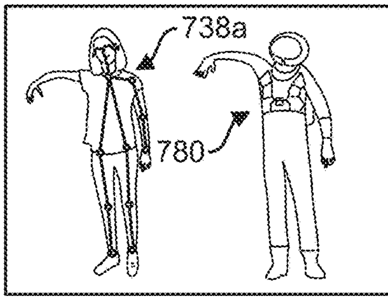


FIG. 7A

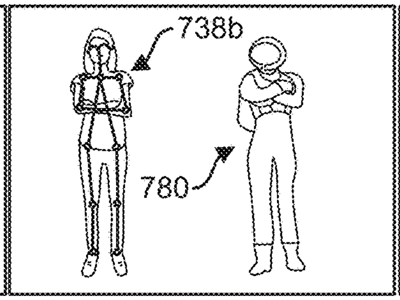


FIG. 7B

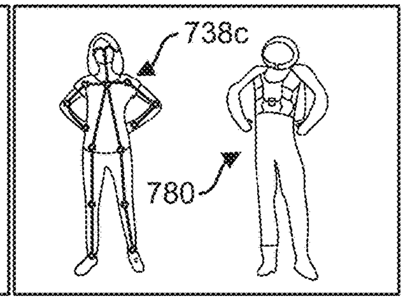


FIG. 7C

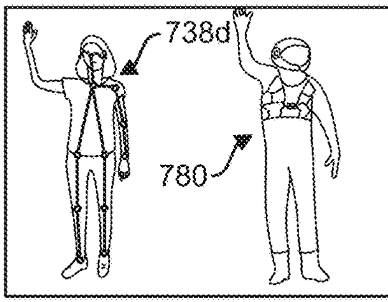


FIG. 7D

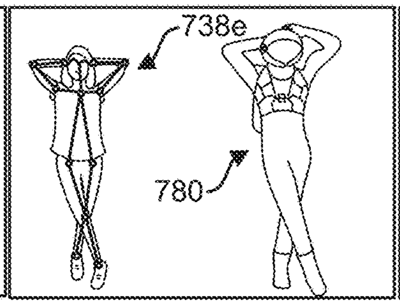


FIG. 7E

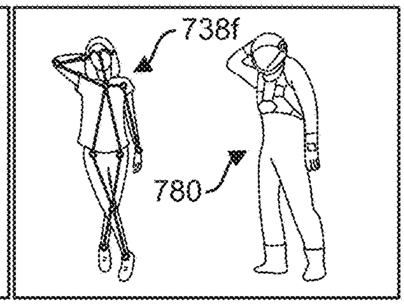


FIG. 7F

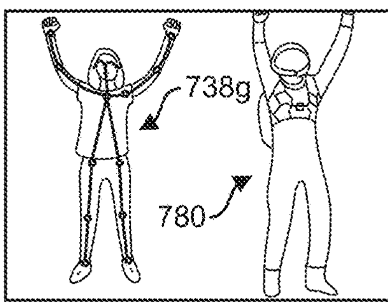


FIG. 7G

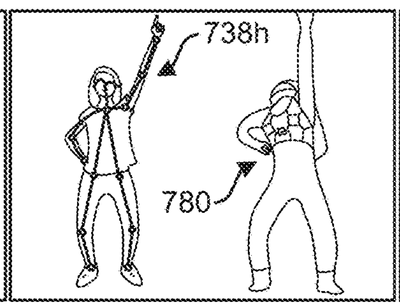


FIG. 7H

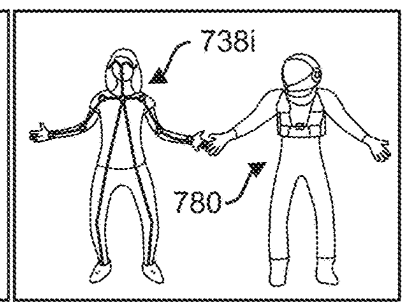


FIG. 7I

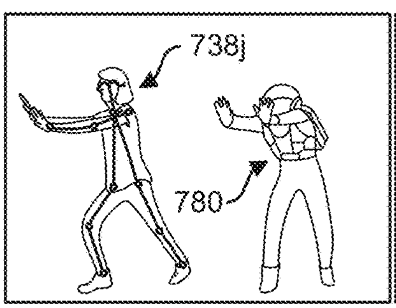


FIG. 7J

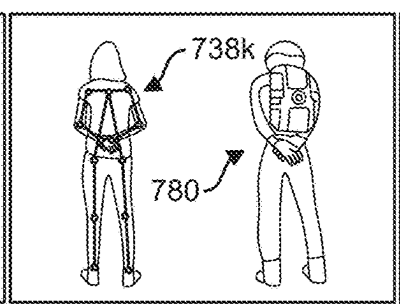


FIG. 7K

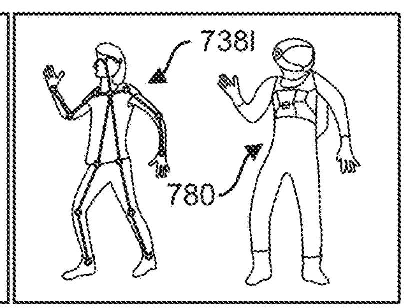


FIG. 7L

AUTOMATED IMAGE AUGMENTATION USING A VIRTUAL CHARACTER

The present application is a Continuation of U.S. application Ser. No. 16/037,745, filed Jul. 17, 2018, which is hereby incorporated by reference in its entirety into the present application.

BACKGROUND

Augmented reality (AR), in which real world objects and/or environments are digitally augmented with virtual imagery, offers more immersive and enjoyable educational or entertainment experiences.

SUMMARY

There are provided systems and methods for performing automated image augmentation using a virtual character, substantially as shown in and/or described in connection with at least one of the figures, and as set forth more completely in the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** shows a diagram of an exemplary system for performing automated image augmentation, according to one implementation;

FIG. **2** shows a more detailed exemplary representation of a remote communication device suitable for use in performing automated image augmentation, in combination with a computer server;

FIG. **3** shows a flowchart presenting an exemplary method for performing automated image augmentation, according to one implementation;

FIG. **4** shows an exemplary two-dimensional (2D) skeleton suitable for use in performing image augmentation using a virtual character, according to one implementation;

FIG. **5** depicts three-dimensional (3D) pose projection based on a 2D skeleton, according to one implementation;

FIGS. **6**A and **6**B show exemplary augmented images each including a virtual character; and

FIGS. **7**A-**7**L show exemplary 3D poses and a virtual character based on the 3D pose shown in each figure.

DETAILED DESCRIPTION

Despite its usefulness in augmenting many inanimate objects, however, digital augmentation of the human body continues to present significant technical challenges. For example, due to the ambiguities associated with depth projection, as well as the variations in human body shapes, three-dimensional (3D) human pose estimation from a red-green-blue (RGB) image is an under-constrained and ambiguous problem.

Although solutions for estimating a human pose using a depth camera have been proposed, they typically require the preliminary generation of a large data set of 3D skeleton poses and depth image pairs. A machine learning model is then trained to map depth to 3D skeletons. In addition to the pre-processing burdens imposed by such approaches, there are the additional disadvantages that a large data set of 3D skeleton poses can be complicated to gather, as well as the possibility that those data sets may not include all of the poses that are useful or relevant to a particular application.

The following description contains specific information pertaining to implementations in the present disclosure. One skilled in the art will recognize that the present disclosure may be implemented in a manner different from that specifically discussed herein. The drawings in the present application and their accompanying detailed description are directed to merely exemplary implementations.

Unless noted otherwise, like or corresponding elements among the figures may be indicated by like or corresponding reference numerals. Moreover, the drawings and illustrations in the present application are generally not to scale, and are not intended to correspond to actual relative dimensions.

FIG. **1** shows a diagram of an exemplary system for performing automated image augmentation. As shown in FIG. **1**, image processing system **100** includes computer server **102** having hardware processor **104**, and system memory **106** implemented as a non-transitory storage device. According to the present exemplary implementation, system memory **106** stores image augmentation software code **110**, one or both of 3D poses library **112** and 3D shapes library **114**, and 2D pose estimation module **108**.

As further shown in FIG. **1**, computer server **102** is implemented within a use environment including network **120**, communication device **140** remote from computer server **102** (hereinafter "remote communication device **140**") including display **142**, first user **124** (hereinafter "user **124**") utilizing remote communication device **140**, and second user or subject **125** (hereinafter "subject **125**"). Also shown in FIG. **1** are network communication links **122** communicatively coupling remote communication device **140** to computer server **102** via network **120**, image **130**, and 2D pose data **132** generated by 2D pose estimation module **108**, as well as augmented image **190** or 3D shape and/or 3D pose **138** output by image augmentation software code **110**.

It is noted that, although the present application refers to image augmentation software code **110**, one or both of 3D poses library **112** and 3D shapes library **114**, and 2D pose estimation module **108** as being stored in system memory **106** for conceptual clarity, more generally, system memory **106** may take the form of any computer-readable non-transitory storage medium. The expression "computer-readable non-transitory storage medium," as used in the present application, refers to any medium, excluding a carrier wave or other transitory signal that provides instructions to a hardware processor of a computing platform, such as hardware processor **104** of computer server **102**. Thus, a computer-readable non-transitory medium may correspond to various types of media, such as volatile media and non-volatile media, for example. Volatile media may include dynamic memory, such as dynamic random access memory (dynamic RAM), while non-volatile memory may include optical, magnetic, or electrostatic storage devices. Common forms of computer-readable non-transitory media include, for example, optical discs, RAM, programmable read-only memory (PROM), erasable PROM (EPROM), and FLASH memory.

It is further noted that although FIG. **1** depicts image augmentation software code **110**, one or both of 3D poses library **112** and 3D shapes library **114**, and 2D pose estimation module **108** as being co-located in system memory **106**, that representation is also provided merely as an aid to conceptual clarity. More generally, image processing system **100** may include one or more computing platforms corresponding to computer server **102** and/or remote communication device **140**, which may be co-located, or may form an interactively linked but distributed system, such as a cloud based system, for instance.

As a result, hardware processor **104** and system memory **106** may correspond to distributed processor and memory resources within image processing system **100**. Thus, it is to be understood that image augmentation software code **110**, one or both of 3D poses library **112** and 3D shapes library **114**, and 2D pose estimation module **108** may be stored and/or executed using the distributed memory and/or processor resources of image processing system **100**.

Image processing system **100** provides an automated image processing solution for augmenting an image portraying a human being, with a virtual character. Image processing system **100** does so at least in part by providing the image as an input to a two-dimensional (2D) pose estimation module and receiving a 2D pose data generated by the 2D pose estimation module based on the image. Image processing system **100** further identifies one or more of a three-dimensional (3D) pose and a 3D shape corresponding to the human portrayal (hereinafter "human image") based on the 2D pose data. The identified 3D shape and/or 3D pose can then be used to size and/or pose a virtual character for inclusion in an augmented image including the human image and/or the virtual character.

In one implementation of image processing system **100**, the virtual character may appear in the augmented image beside the human image, and may assume a posture or pose that substantially reproduces the pose of the human image. In another implementation, the virtual character may partially overlap the human image, such as by appearing to have an arm encircling the shoulders or waist of the human image, for example. In yet another implementation, the virtual character may substantially overlap and obscure the human image so as to appear to be worn as a costume by the human image. As a result, image processing system **100** advantageously provides a fully automated solution for generating augmented self-images, such as so called "selfies," or other images for a user.

For example, in some implementations, user **124** may utilize remote communication device **140** to obtain a selfie, which may be augmented by image processing system **100**. Alternatively, or in addition, user **124** may utilize remote communication device **140** to obtain an image of another person, such as subject **125**, which may be augmented by image processing system **100**. These implementations and more are discussed in more detail below.

Turning to the implementation shown in FIG. **1**, user **124** may utilize remote communication device **140** to interact with computer server **102** over network **120**. In one such implementation, computer server **102** may correspond to one or more web servers, accessible over a packet-switched network such as the Internet, for example. Alternatively, computer server **102** may correspond to one or more computer servers supporting a local area network (LAN), or included in another type of limited distribution network.

Although remote communication device **140** is shown as a mobile device in the form of a smartphone or tablet computer in FIG. **1**, that representation is also provided merely as an example. More generally, remote communication device **140** may be any suitable mobile or stationary computing device or system remote from computer server **102** and capable of performing data processing sufficient to provide a user interface, support connections to network **120**, and implement the functionality ascribed to remote communication device **140** herein. For example, in other implementations, remote communication device **140** may take the form of a laptop computer, or a photo booth in a theme park or other entertainment venue, for example. In one implementation, user **124** may utilize remote communication device **140** to interact with computer server **102** to use image augmentation software code **110**, executed by hardware processor **104**, to produce 3D shape and/or 3D pose **138** for augmenting image **130**.

It is noted that, in various implementations, 3D shape and/or 3D pose **138**, when generated using image augmentation software code **110**, may be stored in system memory **106** and/or may be copied to non-volatile storage. Alternatively, or in addition, as shown in FIG. **1**, 3D shape and/or 3D pose **138** may be sent to remote communication device **140** including display **142**, for example by being transferred via network communication links **122** of network **120**. It is further noted that display **142** may be implemented as a liquid crystal display (LCD), a light-emitting diode (LED) display, an organic light-emitting diode (OLED) display, or any other suitable display screen that performs a physical transformation of signals to light.

FIG. **2** shows a more detailed representation of exemplary remote communication device **240** in combination with computer server **202**. As shown in FIG. **2**, remote communication device **240** is communicatively coupled to computer server **202** over network communication link **222**. Computer server **202** includes hardware processor **204**, and system memory **206** storing image augmentation software code **210***a*, one or both of 3D poses library **212***a* and 3D shapes library **214***a*, and 2D pose estimation module **208**.

As further shown in FIG. **2**, remote communication device **240** includes hardware processor **244**, system memory **246** implemented as a non-transitory storage device storing image augmentation software code **210***b* and one or both of 3D poses library **212***b* and 3D shapes library **214***b*. As also shown in FIG. **2**, remote communication device **240** includes transceiver **252**, camera **254**, and display **242** receiving augmented image **290** from image augmentation software code **210***b*.

Network communication link **222** and computer server **202** having hardware processor **204** and system memory **206**, correspond in general to network communication link **122** and computer server **102** having hardware processor **104** and system memory **106**, in FIG. **1**. In addition, image augmentation software code **210***a*, 3D poses library **212***a*, 3D shapes library **214***a*, and 2D pose estimation module **208**, in FIG. **2**, correspond respectively in general to image augmentation software code **110**, 3D poses library **112**, 3D shapes library **114**, and 2D pose estimation module **108**, in FIG. **1**. In other words, image augmentation software code **210***a*, 3D poses library **212***a*, 3D shapes library **214***a*, and 2D pose estimation module **208** may share any of the characteristics attributed to respectively corresponding image augmentation software code **110**, 3D poses library **112**, 3D shapes library **114**, and 2D pose estimation module **108** by the present disclosure, and vice versa. It is also noted that augmented image **290** and 3D shape and/or 3D pose **238**, in FIG. **2**, correspond respectively in general to augmented image **190** and 3D shape and/or 3D pose **138**, in FIG. **1**.

Remote communication device **240** and display **242** correspond in general to remote communication device **140** and display **142**, in FIG. **1**, and those corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure. Thus, like remote communication device **140**, remote communication device **240** may take the form of a smartphone, tablet or laptop computer, or a photo booth in a theme park or other entertainment venue. In addition, and although not shown in FIG. **1**, remote communication device **140** may include features corresponding to hardware processor **244**, transceiver **252**, camera **254**, and system memory **246** storing image augmentation software code **210***b* and one or both of

3D poses library **212***b* and 3D shapes library **214***b*. Moreover, like display **142**, display **242** may be implemented as an LCD, an LED display, an OLED display, or any other suitable display screen that performs a physical transformation of signals to light.

With respect to image augmentation software code **210***b*, 3D poses library **212***b*, and 3D shapes library **214***b*, it is noted that in some implementations, image augmentation software code **210***b* may be an application providing a user interface for exchanging data, such as data corresponding to image **130** and augmented image **190**/**290** or 3D shape and/or 3D pose **138**/**238** with computer server **102**/**202**. In those implementations, system memory **246** of remote communication device **140**/**240** may not store 3D poses library **212***b* or 3D shapes library **214***b*.

However, in other implementations, image augmentation software code **210***b* may include all of the features of image augmentation software code **110**/**210***a*, and may be capable of executing all of the same functionality. That is to say, in some implementations, image augmentation software code **210***b* corresponds to image augmentation software code **110**/**210***a* and may share any of the features and perform any of the processes attributed to those corresponding features by the present disclosure.

Furthermore, and as shown in FIG. **2**, in implementations in which image augmentation software code **210***b* corresponds to image augmentation software code **110**/**210***a*, one or both of 3D poses library **212***b* and 3D shapes library **214***b* may be stored locally on system memory **246**. It is also noted that, when present in system memory **246** of remote communication device **240**, 3D poses library **212***b* and 3D shapes library **214***b* correspond respectively in general to 3D poses library **112**/**212***a* and 3D shapes library **114**/**214***b*, and may share any of the characteristics attributed to those corresponding features by the present disclosure.

According to the exemplary implementation shown in FIG. **2**, image augmentation software code **210***b* and one or both of 3D poses library **212***b* and 3D shapes library **214***b* are located in system memory **246**, having been received via network communication link **122**/**222**, either from computer server **102**/**202** or an authorized third party source of image augmentation software code **210***b* and one or both of 3D poses library **212***b* and 3D shapes library **214***b*. In one implementation, network communication link **122**/**222** corresponds to transfer of image augmentation software code **210***b* and one or both of 3D poses library **212***b* and 3D shapes library **214***b* over a packet-switched network, for example. Once transferred, for instance by being downloaded over network communication link **122**/**222**, image augmentation software code **210***b* and one or both of 3D poses library **212***b* and 3D shapes library **214***b* may be persistently stored in device memory **246**, and image augmentation software code **210***b* may be executed on remote communication device **140**/**240** by hardware processor **244**.

Hardware processor **244** may be the central processing unit (CPU) for remote communication device **140**/**240**, for example, in which role hardware processor **244** runs the operating system for remote communication device **140**/**240** and executes image augmentation software code **210***b*. As noted above, in some implementations, remote communication device **140**/**240** can utilize image augmentation software code **210***b* as a user interface with computer server **102**/**202** for providing image **130** to image augmentation software code **110**/**210***a*, and for receiving augmented image **190**/**290** or 3D shape and/or 3D pose **138**/**238** from image augmentation software code **110**/**210***a*.

However, in other implementations, remote communication device **140**/**240** can utilize image augmentation software code **210***b* to interact with computer server **102**/**202** by providing image **130** to 2D pose estimation module **108**/**208**, and may receive 2D pose data **132** generated by 2D pose estimation module **108**/**208** via network **120**. In those latter implementations, image augmentation software code **210***b* may further identify 3D shape and/or 3D pose **138**/**238** on remote communication device **140**/**240**, and may use 3D shape and/or 3D pose **138**/**238** to produce augmented image **190**/**290**. Furthermore, in those implementations, hardware processor **244** may execute image augmentation software code **210***b* to render augmented image **190**/**290** on display **142**/**242**.

The functionality of image processing system **100** will be further described by reference to FIG. **3**. FIG. **3** shows flowchart **360** presenting an exemplary method for performing automated image augmentation, according to one implementation. With respect to the method outlined in FIG. **3**, it is noted that certain details and features have been left out of flowchart **360** in order not to obscure the discussion of the inventive features in the present application. It is further noted that the feature "computer server **102**/**202**" described in detail above will hereinafter be referred to as "computing platform **102**/**202**," while the feature "remote communication device **140**/**240**" will hereinafter be referred to as "remote computing platform **140**/**240**."

Referring to FIG. **3** in combination with FIGS. **1** and **2**, flowchart **360** begins with providing image **130** as an input to 2D pose estimation module **108**/**208** (action **362**). Image **130** may be a red-green-blue (RGB) image obtained by a digital camera, such as a digital still image camera for example. Alternatively, image **130** may be an RGB image taken from a video clip obtained by a digital video camera. In one implementation, image **130** may be a single monocular image including a human image portraying a human body in a particular posture or pose, for example.

In some implementations, hardware processor **244** of remote computing platform **140**/**240** may execute image augmentation software code **210***b* to obtain image **130** using camera **254**. Thus, camera **254** may be an RGB camera configured to obtain still or video digital images.

In some implementations, image **130** may be transmitted by remote computing platform **140**/**240**, using transceiver **252**, to computing platform **102**/**202** via network **120** and network communication links **122**/**222**. In those implementations, image **130** may be received by image augmentation software code **110**/**210***a*, executed by hardware processor **104**/**204** of computing platform **102**/**202**. However, in other implementations, image **130** may be received from camera **254** by image augmentation software code **210***b*, executed by hardware processor **244** of remote computing platform **140**/**240**.

In implementations in which image **130** is received by image augmentation software code **110**/**210***a* stored in system memory **106**/**206** also storing 2D pose estimation module **108**/**208**, providing image **130** in action **362** may be performed as a local data transfer within system memory **106**/**206** of computing platform **102**/**202**, as shown in FIG. **1**. In those implementations, image **130** may be provided to 2D pose estimation module **108**/**208** by image augmentation software code **110**/**210***a*, executed by hardware processor **104**/**204** of computing platform **102**/**202**.

However, as noted above, in some implementations, image **130** is received by image augmentation software code **210***b* stored in system memory **246** of remote computing platform **140**/**240**. In those implementations, remote computing platform is remote from 2D pose estimation module **108/208**. Nevertheless, and as shown by FIG. **1**, 2D pose estimation module **108/208** may be communicatively coupled to image augmentation software code **210***b* via network **120** and network communication links **122/222**. In those implementations, image **130** may be provided to 2D pose estimation module **108/208** via network **120** by image augmentation software code **210***b*, executed by hardware processor **244** of remote computing platform **140/240**, and using transceiver **252**.

Flowchart **360** continues with receiving from 2D pose estimation module **108/208**, 2D pose data **132** generated based on image **130** (action **364**), 2D pose estimation module **108/208** may be configured to run a deep neural network, as known in the art, which takes image **130** as input, and returns 2D pose data **132** as a list of joint positions y_i together with a confidence value c_i for each joint position y_i. For example, when image **130** includes an image of a partially visible human body, a low confidence value will result for joints outside of view. The deep neural network of 2D pose estimation module **108/208** may have been previously trained over a large data set of hand annotated images, for example, but may be implemented so as to generate pose data **132** based on image **130** in an automated process.

Referring to FIG. **4**, FIG. **4** shows 2D skeleton **470** described by 2D pose data **132** generated based on a human image included in image **130**. 2D skeleton **470** has a set of joint names corresponding to of joint positions y_i, in this exemplary case thirteen, that can subsequently be associated to 3D joints of a 3D virtual character.

In implementations in which image **130** is provided to 2D pose estimation module **108/208** by image augmentation software code **110/210***a*, receiving 2D pose data **132** may be performed as a local data transfer within system memory **106/206** of computing platform **102/202**, as shown in FIG. **1**. In those implementations, 2D pose data **132** may be received from 2D pose estimation module **108/208** by image augmentation software code **110/210***a*, executed by hardware processor **104/204** of computing platform **102/202**.

However, in implementations in which image **130** is provided to 2D pose estimation module **108/208** remote from remote computing platform **140/240** by image augmentation software code **210***b*, 2D pose data **132** may be received via network **120** and network communication links **122/222**. As shown in FIG. **1**, in those implementations, 2D pose data **132** may be received from remote 2D pose estimation module **108/208** via network **120** by image augmentation software code **210***b*, executed by hardware processor **244** of remote computing platform **140/240**, and using transceiver **252**.

Flowchart **360** continues with identifying 3D shape and/or 3D pose **138/238** corresponding to image **130**, based on 2D pose data **132**, where 3D shape and/or 3D pose **138/238** is identified using an optimization algorithm applied to 2D pose data **132** and one or both of 3D poses library **112/212***a*/**212***b* and 3D shapes library **114/214***a*/**214***b* (action **366**). It is noted that the description below refers specifically to the identification of a 3D pose **138/238** corresponding to image **130**. However and as further discussed briefly below, the present approach may be readily extended for identification of a 3D shape **138/238** corresponding to image **130**.

When identifying 3D pose **138/238** corresponding to image **130**, it may be advantageous or desirable to include only a relatively small number of 3D poses in 3D poses library **112/212***a*/**212***b*. For example, in one implementation, 3D poses library **112/212***a*/**212***b* may include twelve 3D poses. 3D pose **138/238** corresponding to image **130** may be identified by projecting 2D skeleton **470** described by 2D pose data **132** onto the 3D pose space defined by the 3D poses included in 3D poses library **112/212***a*/**212***b* via local optimization. For example, for each 3D pose in 3D poses library **112/212***a*/**212***b*, the present solution optimizes for the rigid transformation that will bring the 3D poses in 3D poses library **112/212***a*/**212***b* closest to the projection of 2D skeleton **470**, in terms joint positions and bone direction similarity.

Referring to FIG. **5**, FIG. **5** shows diagram **500** depicting 3D pose projection based on a 2D skeleton, according to one implementation. FIG. **5** shows projection of 2D skeleton **570** and human image **568** included in image **130**. FIG. **5** also shows virtual character **580**, rotational axis y **582**, and mutually orthogonal translational axes x **584** and z **586**. It is noted that 2D skeleton **570** corresponds in general to 2D skeleton **470** and may share any of the characteristics attributed to that corresponding feature by the present disclosure.

Formally, for each pose $X^k=\{x_i\}^k$ defined as a set of joint positions x_i, we optimize for a reduced rigid transformation M composed of a rotation around the y axis **582** (R_y), and translations along the x axis **584** (T_x) and z axis **586** (T_z) resulting in $M=T_zT_xR_y$. The rigid transformation M minimizes the similarity cost between the 3D projected joint positions P, M, x_i, and the 2D joint positions y_i, where P is a view and projection transformation of the camera used to obtain image **130**. Finally, we analyze all the optimal transformation and pose pairs $<X^k, M>$, and identify the one that has the smallest cost value, resulting in the following optimization problem:

$$X^*, M^* = \underset{<X^k, M>}{\arg\min}\ \underset{M}{\min} \sum_i \|y_i - PMx_i\|^2 \quad \text{(Equation 1)}$$

The internal optimization for the transformation M is solved using gradient-based optimization along numerical derivatives. This requires initializing the 3D pose front facing the camera as to ensure convergence towards a sensible solution.

In order to incorporate an apparently 3D virtual character into augmented image **190/290**, a view and perspective matrix (P) of the camera used to obtain image **130** is needed. Where image **130** is obtained using camera **254** of remote computing platform **140/240** implemented as part of a mobile communication device, for example, the perspective matrix P is given by the mobile communication device, while a conventional marker-based technology can be utilized to recognize and track the transformations of camera **254**. In one implementation, a real world marker that is approximately the size of a person can be included in image **130**. When camera **254** obtains image **130**, image **130** contains the marker, which may be used to estimate the orientation and position of camera **254**.

The 3D pose identified using Equation 1 may be initialized to approximately fit inside the bounding box of the marker. The optimization algorithm adjusts the virtual character's depth translation to substantially match the of 2D skeleton **470/570**. If the virtual character is to be smaller, (e.g. a dwarf), final 3D pose **138/238** can be scaled back to its original size at the end of the optimization process.

The description of action **364** provided above refers to identification of 3D pose **138/238** corresponding to image **130**. For example, where 3D poses library **112/212***a*/**212***b* includes twelve 3D poses, an optimization algorithm for solving Equation 1 may be applied to every one of the twelve 3D poses in 3D poses library **112**/**212***a*/**212***b*, i.e., the entirety of 3D poses library **112**/**212***a*/**212***b*.

By analogy, when action **364** includes identification of 3D shape **138**/**238** corresponding to image **130**, an analogous optimization algorithm can be applied to every combination of every 3D pose in 3D poses library **112**/**212***a*/**212***b* with every 3D shape exemplar in 3D shapes library **114**/**214***a*/**214***b*. For example, where 3D poses library **112**/**212***a*/**212***b* includes twelve 3D poses and 3D shapes library **114**/**214***a*/**214***b* includes five 3D shape exemplars, an optimization algorithm for solving an optimization problem analogous to Equation 1 may be applied to each of the sixty combinations (12×5) of the twelve 3D poses and five 3D shape exemplars, i.e., the entirety of 3D poses library **112**/**212***a*/**212***b* and 3D shapes library **114**/**214***a*/**214***b*.

In implementations in which 2D pose data **132** is received from 2D pose estimation module **108**/**208** by image augmentation software code **110**/**210***a*, identification of 3D shape and/or 3D pose **138**/**238** may be performed by image augmentation software code **110**/**210***a*, executed by hardware processor **104**/**204** of computing platform **102**/**202**. However, in implementations in which 2D pose data is received from 2D pose estimation module **108**/**208** by image augmentation software code **210***b* on remote computing platform **140**/**240**, identification of 3D shape and/or 3D pose **138**/**238** may be performed by image augmentation software code **210***b*, executed by hardware processor **244** of remote computing platform **140**/**240**.

In some implementation, the method outlined by flowchart **360** can conclude with action **366** described above. However, in other implementations, flowchart **360** can continue with outputting 3D shape and/or 3D pose **138**/**238** to render augmented image **190**/**290** on display **142**/**242** (action **368**). In implementations in which 3D shape and/or 3D pose **138**/**238** is identified by image augmentation software code **110**/**210***a*, image augmentation software code **110**/**210***a* may be further executed by hardware processor **104**/**204** of computing platform **102**/**202** to output 3D shape and/or 3D pose **138**/**238** by transmitting 3D shape and/or 3D pose **138**/**238** to remote computing platform **140**/**240** via network **120** and network communication links **122**/**222**.

However, in implementations in which 3D shape and/or 3D pose **138**/**238** is identified by image augmentation software code **210***b*, image augmentation software code **210***b* may output 3D shape and/or 3D pose **138**/**238** as part of augmented image **190**/**290**. In those implementations, for example, hardware processor **244** may execute image augmentation software code **210***b* to use 3D shape and/or 3D pose to produce augmented image **190**/**290**, and to render augmented image **190**/**290** on display **142**/**242**. As yet another alternative, in one implementation, image augmentation software code **110**/**210***a* of computing platform **102**/**202** may output 3D shape and/or 3D pose **138**/**238** as part of augmented image **190**/**290**, and may transmit augmented image **190**/**290** to remote computing platform **140**/**240** for rendering on display **142**/**242**.

It is noted that in the various implementations described above, augmented image **190**/**290** can be rendered on display **142**/**242** in real-time with respect to receipt of image **130** by image augmentation software code **110**/**210***a* or **210***b*. For example, in some implementations, a time lapse between receiving image **130** by image augmentation software code **110**/**210***a* or **210***b* and rendering augmented image **190**/**290** on display **142**/**242** may be less than five seconds, such as two to three seconds.

FIG. **6**A shows exemplary augmented image **690**A including human image **668** and virtual character **680** generated based on 3D shape and/or 3D pose **138**/**238**, according to one implementation, while FIG. **6**B shows exemplary augmented image **690**B including human image **668** and virtual character **680** generated based on another 3D shape and/or 3D pose **138**/**238**. Also included in FIGS. **6**A and **6**B is marker **688** that can be utilized to recognize and track the transformations of the camera obtaining image **130** on which augmented image **190**/**290** is based, as described above by reference to action **364**.

It is noted that augmented images **690**A and **690**B correspond in general to augmented image **190**/**290**, in FIG. **2**. Consequently, augmented image **190**/**290** may share any of the characteristics attributed to augmented images **690**A and/or **690**B, while augmented images **690**A and **690**B may share any of the characteristics attributed to augmented image **190**/**290** by the present disclosure. It is further noted that human image **668** corresponds to human image **568**, in FIG. **5**, and those corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure. Moreover, virtual character **680** corresponds to virtual character **580**, and those corresponding features may share any of the characteristics attributed to either corresponding feature by the present disclosure.

As shown in FIGS. **6**A and **6**B, where image **130** includes human image **568**/**668**, augmented image **190**/**290**/**690**A/**690**B includes human image **568**/**668** and virtual character **580**/**680** generated based on 3D shape and/or 3D pose **138**/**238**. In some implementations, virtual character **580**/**680** may be generated based on 3D pose **138**/**238** alone. In those implementations, and as shown in FIGS. **6**A and **6**B, virtual character **580**/**680** and human image **568**/**668** are typically non-overlapping in augmented image **190**/**290**/**690**A/**690**B. For example, and as further shown in FIGS. **6**A and **6**B, virtual character **580**/**680** appears in augmented image **190**/**290**/**690**A/**690**B beside human image **568**/**668**, and assumes a pose that substantially reproduces the pose of human image **568**/**668**.

However, in implementations in which augmented image **190**/**290**/**690**A/**690**B includes human image **568**/**668** and virtual character **580**/**680** generated based on 3D shape and 3D pose **138**/**238**, virtual character **580**/**680** may at least partially overlap human image **568**/**668**. For example, in some implementations, virtual character **580**/**680** may partially overlap human image **568**/**668** by appearing to have an arm encircling the shoulders or waist of human image **568**/**668**. In yet another implementation, virtual character **580**/**680** may substantially overlap and obscure human image **568**/**668** so as to appear to be worn as a costume or suit by human image **568**/**668**.

In some implementations, the shape of the person corresponding to human image **568**/**668** can be estimated from a humanoid. That 3D geometry estimation may then be used to support partial occlusions as well as casting approximate shadows from human image **568**/**668** to virtual character **580**/**680**. Thus, in some implementations, hardware processor **104**/**204** may further execute image augmentation software code **110**/**210***a* to estimate a 3D shape corresponding to human image **568**/**668** and utilize that 3D shape to generate one or more of a partial occlusion of virtual character **580**/**680** by human image **568**/**668** and a shadow or shadows cast from human image **568**/**668** to virtual character **580**/**680**. Moreover, in other implementations, hardware processor **244** may further execute image augmentation software code **210***b* to estimate the 3D shape corresponding to human image **568**/**668** and utilize that 3D shape to generate one or more of the partial occlusion of virtual character **580/680** by human image **568/668** and the shadow(s) cast from human image **568/668** to virtual character **580/680**.

FIGS. **7**A-**7**L show exemplary 3D poses **738** and virtual character **780** based on the 3D pose shown in each figure. It is noted that 3D poses **738** correspond in general to 3D shapes and/or 3D poses **138/238**, in FIGS. **1** and **2**, and those corresponding features may share the characteristics attributed to any of the corresponding features by the present disclosure. In addition, virtual character **780** corresponds in general to virtual character **580/680** in FIGS. **5**, **6**A, and **6**B, and those corresponding features may share the characteristics attributed to any of the corresponding features by the present disclosure.

FIG. **7**A shows front facing 3D pose **738***a* with legs together, left arm lowered at left side, and right arm raised below head with elbow bent. Also shown in FIG. **7**A is virtual character **780** assuming substantially the same pose with legs together, left arm lowered at left side, and right arm raised below head with elbow bent.

FIG. **7**B shows front facing 3D pose **738***b* with legs together and arms crossed over chest. Also shown in FIG. **7**B is virtual character **780** assuming substantially the same pose with legs together and arms crossed over chest.

FIG. **7**C shows front facing 3D pose **738***c* with legs slightly spread, elbows bent, and hands on hips. Also shown in FIG. **7**B is virtual character **780** assuming substantially the same pose with legs slightly spread, elbows bent, and hands on hips.

FIG. **7**D shows front facing 3D pose **738***d* with legs together, left arm lowered at left side, and right arm raised above head with elbow bent. Also shown in FIG. **7**D is virtual character **780** assuming substantially the same pose with legs together, left arm lowered at left side, and right arm raised above head with elbow bent.

FIG. **7**E shows front facing 3D pose **738***e* with right leg crossed in front of left leg and hands clasped behind head. Also shown in FIG. **7**E is virtual character **780** assuming substantially the same pose with right leg crossed in front of left leg and hands clasped behind head.

FIG. **7**F shows front facing 3D pose **738***f* with right leg crossed in front of left leg, left arm lowered at left side, and right hand behind head. Also shown in FIG. **7**F is virtual character **780** assuming substantially the same pose with right leg crossed in front of left leg, left arm lowered at left side, and right hand behind head.

FIG. **7**G shows front facing 3D pose **738***g* with legs slightly spread and both arms raised above head. Also shown in FIG. **7**G is virtual character **780** assuming substantially the same pose with legs slightly spread and both arms raised above head.

FIG. **7**H shows front facing 3D pose **738***h* with legs slightly spread, knees slightly bent, right hand on right hip, and left arm raised above head. Also shown in FIG. **7**H is virtual character **780** assuming substantially the same pose with legs slightly spread, knees slightly bent, right hand on right hip, and left arm raised above head.

FIG. **7**I shows front facing 3D pose **738***i* with legs slightly spread, knees slightly bent, and arms extended laterally to the sides with elbows bent. Also shown in FIG. **7**I is virtual character **780** assuming substantially the same pose with legs slightly spread, knees slightly bent, and arms extended laterally to the sides with elbows bent.

FIG. **7**J shows side facing 3D pose **738***j* with bent right leg lunging forward of left leg and both arms extended forward. Also shown in FIG. **7**J is virtual character **780** assuming substantially the same pose with bent right leg stepping forward of left leg and both arms extended forward.

FIG. **7**K shows rear facing 3D pose **738***k* with legs straight and hands clasped at the small of the back. Also shown in FIG. **7**K is virtual character **780** assuming substantially the same pose with legs straight and hands clasped at the small of the back.

FIG. **7**L shows side facing 3D pose **738***l* with bent right leg stepping forward of left leg, right arm raised with elbow bent, and left arm extended backward with elbow bent. Also shown in FIG. **7**L is virtual character **780** assuming substantially the same pose with bent right leg stepping forward of left leg, right arm raised with elbow bent, and left arm extended backward with elbow bent.

It is noted that, in some implementations, 3D poses **738***a* through **738***l* may be stored in 3D poses library **112/212***a***/212***b*. Moreover, in some implementations, 3D poses **738***a* through **738***l* may correspond respectively to substantially all 3D poses stored in 3D poses library **112/212***a***/212***b*.

From the above description it is manifest that various techniques can be used for implementing the concepts described in the present application without departing from the scope of those concepts. Moreover, while the concepts have been described with specific reference to certain implementations, a person of ordinary skill in the art would recognize that changes can be made in form and detail without departing from the scope of those concepts. As such, the described implementations are to be considered in all respects as illustrative and not restrictive. It should also be understood that the present application is not limited to the particular implementations described herein, but many rearrangements, modifications, and substitutions are possible without departing from the scope of the present disclosure.

What is claimed is:

1. An image processing system comprising:

a computing platform including a hardware processor and a system memory;

the system memory storing an image augmentation software code; and

a two-dimensional (2D) pose estimation software having computing instructions;

wherein the hardware processor is configured to execute the image augmentation software code to:

provide an image as an input to the 2D pose estimation software, the pose estimation software including a deep neural network trained over a data set of manually annotated images to receive the image as input and output 2D pose data including a plurality of joint positions and a respective confidence value for each of the plurality of joint positions;

receive from the 2D pose estimation software, the 2D pose data generated based on the image;

generate a virtual character;

estimate a three-dimensional (3D) body shape corresponding to the image based on a rigid transformation of one of a plurality of predetermined 3D body shape exemplars that brings the one of the plurality of predetermined 3D body shape exemplars closest, in terms of joint positions and bone direction similarity, to a 2D pose described by the 2D pose data; and

utilize the 3D body shape to generate a shadow cast from the image to the virtual character.

2. The image processing system of claim **1**, wherein the hardware processor is further configured to execute the image augmentation software code to render, on a display, an augmented image including the image, the virtual character, and the shadow cast from the image to the virtual character.

3. The image processing system of claim 1, wherein the image is a single image.

4. The image processing system of claim 1, wherein the computing platform is part of a device remote from the 2D pose estimation software, the device further comprising a display and a camera, and wherein the hardware processor is further configured to execute the image augmentation software code to obtain the image using the camera.

5. The image processing system of claim 1, wherein the hardware processor is further configured to execute the image augmentation software code to:

render, on a display, an augmented image in which the virtual character appears as a copy of at least a portion of the image and is wearing a suit or a costume.

6. The image processing system of claim 1, wherein the system memory further stores a 3D poses library including a plurality of 3D poses, wherein the hardware processor is further configured to execute the image augmentation software code to:

project the 2D pose data onto a 3D pose space defined by the plurality of 3D poses in the 3D poses library to identify a 3D pose corresponding to the image;

wherein the virtual character is generated having the identified 3D pose.

7. The image processing system of claim 1, wherein the system memory further stores a 3D body shapes library including the plurality of predetermined 3D body shape exemplars, and wherein the hardware processor is further configured to execute the image augmentation software code to:

identify the 3D body shape corresponding to the image, using the plurality of predetermined 3D body shape exemplars in the 3D body shapes library.

8. The image processing system of claim 7, wherein the hardware processor is further configured to execute the image augmentation software code to:

utilize the 3D body shape to generate a partial occlusion of the virtual character by the image.

9. The image processing system of claim 6, wherein the image includes a subject, and wherein the hardware processor is further configured to execute the image augmentation software code to:

render, on a display, the subject beside the virtual character, wherein both the subject and the virtual character have the identified 3D pose.

10. A computer-implemented method for generating a virtual character, the method comprising:

providing an image as an input to a two-dimensional (2D) pose estimation software having computing instructions, the pose estimation software including a deep neural network trained over a data set of manually annotated images to receive the image as input and output 2D pose data including a plurality of joint positions and a respective confidence value for each of the plurality of joint positions;

receiving from the 2D pose estimation software, a 2D pose data generated based on the image;

estimating a three-dimensional (3D) body shape corresponding to the image based on a rigid transformation of one of a plurality of predetermined 3D body shape exemplars that brings the one of the plurality of predetermined 3D body shape exemplars closest, in terms of joint positions and bone direction similarity, to a 2D pose described by the 2D pose data; and

utilizing the 3D body shape to generate a shadow cast from the image to the virtual character.

11. The computer-implemented method of claim 10, further comprising:

rendering, on a display, an augmented image including the image, the virtual character, and the shadow cast from the image to the virtual character.

12. The computer-implemented method of claim 10, wherein the image is a single image.

13. The computer-implemented method of claim 10, wherein the image is provided by a device remote from the 2D pose estimation software, the device comprising a display and a camera, the computer-implemented method further comprising:

obtaining the image using the camera.

14. The computer-implemented method of claim 10, the method further comprising:

rendering, on a display, an augmented image in which the virtual character appears as a copy of at least a portion of the image and is wearing a suit or a costume.

15. The computer-implemented method of claim 10, further comprising:

projecting the 2D pose data onto a 3D pose space defined by the plurality of 3D poses in the 3D poses library to identify a 3D pose corresponding to the 3D image;

wherein the virtual character is generated having the identified 3D pose.

16. The computer-implemented method of claim 10, further comprising:

identifying the 3D body shape corresponding to the image, using a 3D body shapes library including the plurality of predetermined 3D body shape exemplars.

17. The computer-implemented method of claim 16, further comprising:

utilizing the 3D body shape to generate a partial occlusion of the virtual character by the image.

18. The computer-implemented method of claim 15, wherein the image includes a subject, the computer-implemented method further comprising:

rendering, on a display, the subject beside the virtual character, wherein both the subject and the virtual character have the identified 3D pose.

19. A computer-implemented method for generating a virtual character, the method comprising:

providing an image of a human as an input to a two-dimensional (2D) pose estimation software having computing instructions, the pose estimation software including a deep neural network trained over a data set of manually annotated images to receive the image as input and output 2D pose data including a plurality of joint positions and a respective confidence value for each of the plurality of joint positions;

receiving from the 2D pose estimation software, a 2D pose data generated based on the image of the human;

identifying a three-dimensional (3D) shape corresponding to the image of the human using the 2D pose data and a 3D shapes library including a plurality of predetermined 3D body shape exemplars, based on a rigid transformation of one of the plurality of predetermined 3D body shape exemplars that brings the one of the plurality of predetermined 3D body shape exemplars closest, in terms of joint positions and bone direction similarity, to a 2D human pose described by the 2D pose data;

generating the virtual character having the identified 3D shape; and utilizing the 3D shape to generate a partial occlusion of the virtual character by the image of the human.

20. The computer-implemented method of claim **19**, further comprising:

rendering, on a display, an augmented image, the virtual character being partially occluded by the image of the human.

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